Cohomological Connections Between the Cost-First Ledger Framework and Aperiodic Tiling Spaces

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Abstract

We develop a formal cohomological framework connecting the discrete potential theory of the cost-first ledger [1] with the topological invariants of Penrose tiling spaces. The ledger framework produces closed 1-cochains on a recognition graph via cycle-closure conditions, which admit scalar potentials by a discrete Poincaré lemma. Independently, the topology of tiling spaces—studied via the Anderson–Putnam complex [3]—yields Čech cohomology groups that classify phason degrees of freedom and shape deformations. We show that the five Ammann bar cochains naturally associated with (and refining) the ledger's edge-ratio postings are pattern-equivariant 1-cocycles on the tiling graph, and that their cohomology classes generate the first Čech cohomology of the Penrose tiling space $\check{H}^1(\Omega_P;\mathbb{Z}) \cong \mathbb{Z}^5$ via the Kellendonk–Putnam pattern-equivariant isomorphism [6]. This identification realizes the ledger's potentials (height functions) as the phason coordinates of the tiling: these potentials exist globally but are not pattern-equivariant, so the corresponding cocycles represent nontrivial classes in pattern-equivariant cohomology while being exact in ordinary cohomology. We discuss implications for the cost-theoretic interpretation of phason strain, the classification of local matching rules via cohomological obstructions, and the extension to higher-dimensional quasicrystalline systems.

Keywords: cohomology of tiling spaces, Anderson–Putnam complex, ledger potential, phason coordinates, aperiodic order, cost-first framework

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1 Introduction

Two independently motivated frameworks assign algebraic invariants to aperiodic structures. On one hand, the *topological theory of tiling spaces* [3, 4, 5] studies the hull Ω of a tiling—the space of all tilings locally indistinguishable from a given one—and computes its Čech cohomology $\check{H}^*(\Omega;\mathbb{Z})$. For Penrose tilings, $\check{H}^1(\Omega_P;\mathbb{Z}) \cong \mathbb{Z}^5$, encoding the five independent

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phason degrees of freedom associated with the five families of Ammann bars [3, 7]. These cohomological invariants classify which deformations of the tiling preserve its local structure (local indistinguishability class) and which do not.

On the other hand, the *cost-first ledger framework* [1] constructs a discrete potential theory on recognition graphs: edge flows arising from atomic recognition events are required to satisfy time-aggregated cycle closure, yielding closed 1-cochains. The discrete Poincaré lemma then produces scalar potentials (unique up to constants) on each connected component. When the recognition graph is instantiated as the adjacency graph of a Penrose tiling, these potentials coincide with the classical Ammann height functions [8, 9], as established in [2].

The purpose of this note is to make the connection between these two constructions precise at the level of *cochain complexes and cohomology groups*. We show that the ledger's locally defined bar-type cocycles (realized concretely as Ammann bar 1-cocycles $\omega_\tau^{(k)}$ on the tiling graph) are pattern-equivariant 1-cocycles, and that their cohomology classes generate the first Čech cohomology of the Penrose tiling space via the Kellendonk–Putnam pattern-equivariant isomorphism [6]. A key subtlety is that these cocycles are *exact* in ordinary (cellular) cohomology—global height functions exist—but *nontrivial* in pattern-equivariant cohomology, since the height functions are not pattern-equivariant. This identifies the ledger's potential-theoretic data with the topological invariants of the tiling.

1.1 Organization

Section 2 reviews the cohomology of tiling spaces via the Anderson–Putnam construction. Section 3 formalizes the ledger's cochain complex. Section 4 constructs the cochain map and proves the main identification theorem. Section 5 interprets the result in terms of phason coordinates and cost-theoretic strain. Section 6 discusses extensions to higher dimensions and other tiling families. Section 8 concludes with open problems.

2 Cohomology of Tiling Spaces

We recall the essential constructions, following [3, 4, 6].

2.1 The tiling hull

Definition 2.1 (Tiling hull)**.** Let $\mathcal{T}$ be a repetitive, aperiodic tiling of $\mathbb{R}^d$ with finite local complexity. The *hull* (or *tiling space*) of $\mathcal{T}$ is

$$\Omega_{\mathcal{T}} := \overline{\{t + \mathcal{T} : t \in \mathbb{R}^d\}},$$

where the closure is taken in the *local topology*: two tilings are close if they agree on a large ball around the origin (after a small translation).

For Penrose tilings, Ω_P is a compact, connected, metrizable space. It admits a free $\mathbb{R}^2$-action by translation. The *canonical transversal* $\Xi_P \subset \Omega_P$ is defined as the set of tilings in Ω_P having a vertex at the origin; it is a Cantor set that serves as a cross-section of the translation action and parametrizes the "internal" (phason) degrees of freedom. (Note: the

orbit space $\Omega_P/\mathbb{R}^2$ is not Hausdorff, since every orbit is dense; Ξ_P is a transversal, not a quotient.)

2.2 The Anderson–Putnam complex

Definition 2.2 (Anderson–Putnam (AP) complex [3])**.** Let $\mathcal{T}$ be a substitution tiling with substitution σ and prototile set $\mathcal{A} = \{t_1, \ldots, t_m\}$. For each prototile t_i, form a CW complex by identifying edges of t_i according to the adjacency rules of $\mathcal{T}$ (collaring). The resulting CW complex Γ_0 is the *zeroth AP approximant*. The substitution σ induces a continuous map $f_\sigma : \Gamma_0 \to \Gamma_0$, and the tiling space is recovered as the inverse limit:

$$\Omega_{\mathcal{T}} \cong \varprojlim(\Gamma_0 \xleftarrow{f_\sigma} \Gamma_0 \xleftarrow{f_\sigma} \cdots).$$

The Čech cohomology of the inverse limit is computed via the direct limit of cellular cohomology:

$$\check{H}^n(\Omega_{\mathcal{T}}; \mathbb{Z}) \cong \varinjlim(H^n(\Gamma_0; \mathbb{Z}) \xrightarrow{f_\sigma^*} H^n(\Gamma_0; \mathbb{Z}) \xrightarrow{f_\sigma^*} \cdots). \tag{1}$$

2.3 Penrose tiling cohomology

For the Penrose tiling, the AP complex Γ_0 can be constructed from the Robinson triangle decomposition (two prototiles: golden triangle T and golden gnomon G, with collared adjacencies). The substitution matrix on 0-cells, 1-cells, and 2-cells of Γ_0 induces maps on the cellular cochain groups.

Theorem 2.3 (Anderson–Putnam [3]; see also [7])**.** *The Čech cohomology of the Penrose tiling space is:*

$$\check{H}^0(\Omega_P; \mathbb{Z}) \cong \mathbb{Z}, \tag{2}$$

$$\check{H}^1(\Omega_P; \mathbb{Z}) \cong \mathbb{Z}^5, \tag{3}$$

$$\check{H}^2(\Omega_P; \mathbb{Z}) \cong \mathbb{Z}^8. \tag{4}$$

The group $\check{H}^1(\Omega_P; \mathbb{Z}) \cong \mathbb{Z}^5$ has a concrete interpretation: its five generators correspond to the five families of *Ammann bars* (parallel lines crossing the tiling at angles $0°, 36°, 72°, 108°, 144°$). Each family defines a height function (1-cocycle) on the tiling, and the five height functions together parametrize the phason degrees of freedom.

3 The Ledger Cochain Complex

We now formalize the algebraic structure underlying the ledger's discrete potential theory, following [1].

3.1 The edge-flow complex

Definition 3.1 (Ledger cochain complex—graph version)**.** Let $G = (V, E)$ be a finite connected graph with edge set E closed under reversal. Define the *ledger cochain complex* as:

$$C_L^0(G) \xrightarrow{\delta_0} C_L^1(G) \longrightarrow 0, \tag{5}$$

where:

- $C_L^0(G) := \{p : V \to \mathbb{Z}\}$ is the group of vertex potentials (0-cochains).
- $C_L^1(G) := \{\omega : E \to \mathbb{Z} \mid \omega(v \to u) = -\omega(u \to v)\}$ is the group of antisymmetric edge flows (1-cochains).
- $\delta_0(p)(u \to v) := p(v) - p(u)$ (discrete gradient).

This is the standard cochain complex of a 1-dimensional CW complex (graph) with coefficients in $\mathbb{Z}$.

When G is the 1-skeleton of a 2-dimensional CW complex (e.g., a tiling), the complex extends naturally:

$$C_L^0(G) \xrightarrow{\delta_0} C_L^1(G) \xrightarrow{\delta_1} C_L^2(G), \tag{6}$$

where $C_L^2(G) := \bigoplus_{F \in \mathcal{F}} \mathbb{Z}$ is the group of face fluxes (2-cochains) indexed by 2-cells (tiles) $\mathcal{F}$, and the coboundary $\delta_1(\omega)(F) := \sum_{e \in \partial F} \omega(e)$ evaluates the flux of ω through each tile boundary.

Remark 3.2 (Cycle flux as a diagnostic). For any cycle γ in G, the *cycle flux* $\sum_{e \in \gamma} \omega(e)$ provides a useful diagnostic: a 1-cochain ω is a cocycle (i.e., $\delta_1(\omega) = 0$ when 2-cells are present) if and only if ω has zero flux through every tile boundary. On a planar tiling, zero flux through every tile implies zero flux through every cycle.

Proposition 3.3 (Complex property)**.** *When 2-cells are present, the ledger cochain complex satisfies $\delta_1 \circ \delta_0 = 0$: the discrete gradient of any potential has zero flux through every tile boundary (and hence through every cycle).*

Proof. For any $p \in C_L^0(G)$ and any cycle $\gamma = (v_0 \to v_1 \to \cdots \to v_n = v_0)$:

$$\sum_{i=0}^{n-1} \delta_0(p)(v_i \to v_{i+1}) = \sum_{i=0}^{n-1} [p(v_{i+1}) - p(v_i)] = p(v_n) - p(v_0) = 0.$$

In particular, $\delta_1(\delta_0(p))(F) = 0$ for every tile F. □

Definition 3.4 (Ledger cohomology)**.** The *n-th ledger cohomology group* is:

$$H_L^n(G) := \ker \delta_n / \operatorname{im} \delta_{n-1}. \tag{7}$$

For the graph complex (5):

- $H_L^0(G) = \ker \delta_0 \cong \mathbb{Z}$ (constant potentials on a connected graph).

- $H^1_L(G) = C^1_L(G)/\operatorname{im}\delta_0 \cong \mathbb{Z}^{b_1(G)}$, where $b_1(G) = |E^+| - |V| + 1$ is the first Betti number and E^+ is a set of representatives for unoriented edges.

For the extended complex (6) on a planar tiling:

- $H^1_L(G) = \ker\delta_1/\operatorname{im}\delta_0 = 0$ (since the plane is simply connected).

Remark 3.5. The ledger cohomology $H^*_L(G)$ is isomorphic to the standard simplicial (or cellular) cohomology $H^*(X;\mathbb{Z})$ of the underlying CW complex X. The key results of [1]—cycle closure (T3), path-independence, and potential existence (T4)—are cohomological statements: the cumulative edge flow $\overline{\Delta}$ is a 1-cocycle ($\delta_1(\overline{\Delta}) = 0$), and on a tree every 1-cocycle is exact ($H^1_L = 0$), with the potential being the primitive.

A crucial point for the tiling application: on the vertex-edge graph $G_{\mathcal{T}}$ of a planar tiling (viewed as the 1-skeleton of a planar CW complex), $H^1_L(G_{\mathcal{T}}) = 0$. Every cocycle is exact, and global potentials (height functions) always exist. Here we are using ordinary cellular cohomology of the (infinite) CW complex with *all* cochains allowed (no compact-support or boundedness condition). The nontrivial topological content of the tiling is captured not by ordinary cohomology but by the *pattern-equivariant* cohomology of Kellendonk–Putnam [6], which restricts to cochains determined by local data (see Section 4).

3.2 The ledger complex on a tiling graph

When the recognition graph G is instantiated as the vertex-edge graph $G_{\mathcal{T}}$ of a Penrose tiling, the ledger cochain complex becomes:

$$C^0_L(G_{\mathcal{T}}) \xrightarrow{\delta_0} C^1_L(G_{\mathcal{T}}) \xrightarrow{\delta_1} C^2_L(G_{\mathcal{T}}). \tag{8}$$

The five families of Ammann bars in a Penrose tiling [8, 9] define five natural 1-cochains. To make this definition purely local (and hence compatible with pattern-equivariance), we assume $\mathcal{T}$ is presented with the standard Ammann-bar decoration (equivalently, with matching-rule decorations that carry the bar information), so that for each oriented edge one can decide from a sufficiently large but finite neighborhood whether the edge crosses a bar of family k and with which transverse sign.

For the k-th Ammann bar family (bars perpendicular to direction $\theta_k = k \cdot 36°$, $k = 0, 1, 2, 3, 4$), choose a positive transverse orientation. The *Ammann bar 1-cochain* $\omega^{(k)}_\tau \in C^1_L(G_{\mathcal{T}})$ is:

$$\omega^{(k)}_\tau(u \to v) := \begin{cases} +1 & \text{if edge } (u,v) \text{ crosses a bar of family } k\text{, with } u \text{ on the negative side,} \\ -1 & \text{if edge } (u,v) \text{ crosses a bar of family } k\text{, with } u \text{ on the positive side,} \\ 0 & \text{if edge } (u,v) \text{ does not cross a bar of family } k. \end{cases} \tag{9}$$

Each $\omega^{(k)}_\tau$ is antisymmetric by construction: $\omega^{(k)}_\tau(v \to u) = -\omega^{(k)}_\tau(u \to v)$.

Remark 3.6 (Connection to edge-ratio postings). In the de Bruijn pentagrid construction [9], each edge of the Penrose tiling belongs to exactly one grid family and crosses exactly one Ammann bar of that family. The edge-ratio postings from [2]—which assign the golden ratio

φ to long edges and 1 to short edges—encode the aggregate length data. The five Ammann bar cochains $\omega_\tau^{(k)}$ refine this to directional data, one cochain per bar family. The long/short distinction corresponds to the spacing structure within each family.

Proposition 3.7. *Each Ammann bar cochain $\omega_\tau^{(k)}$ is a 1-cocycle: $\delta_1(\omega_\tau^{(k)}) = 0$. Equivalently, $\omega_\tau^{(k)}$ has zero flux through the boundary of every tile.*

Proof. Work with a Penrose tiling equipped with its Ammann-bar decoration. Fix k. Along the boundary of any tile F, the decorated bars of family k enter and exit F in matched pairs: each time a bar crosses the boundary, it must cross again to leave the tile, with the opposite transverse orientation. In terms of the cochain $\omega_\tau^{(k)}$, this means that the contributions of the two boundary edges where the bar crosses are $+1$ and -1, cancelling in the sum over ∂F. Summing over all bar segments intersecting the tile, the total flux around ∂F is therefore zero, i.e., $\delta_1(\omega_\tau^{(k)})(F) = \sum_{e\in\partial F} \omega_\tau^{(k)}(e) = 0$. □

Since $\omega_\tau^{(k)}$ is a cocycle and the plane is simply connected, it is exact in ordinary cohomology: there exists a *height function* $h^{(k)} : V \to \mathbb{Z}$ with $\omega_\tau^{(k)} = \delta_0(h^{(k)})$. This is the content of the discrete Poincaré lemma applied to the full planar tiling.

Remark 3.8 (Exactness and the role of pattern-equivariance). On any finite simply connected patch $\mathcal{P} \subset \mathcal{T}$, as well as on the full infinite tiling, $H^1_L(G_\mathcal{T}) = 0$ and every cocycle is exact—global height functions $h^{(k)}$ exist (uniquely up to additive constants). This is because the underlying space (the plane) is contractible.

However, the height functions $h^{(k)}$ grow linearly across the tiling: they take arbitrarily many distinct values and their value at a vertex depends on its global position, not just its local neighborhood. In particular, the $h^{(k)}$ are *not* pattern-equivariant. The Ammann bar cochains $\omega_\tau^{(k)}$, by contrast, *are* pattern-equivariant (their values are determined by purely local data). This creates a nontrivial situation in pattern-equivariant cohomology: the $\omega_\tau^{(k)}$ are cocycles in the pattern-equivariant subcomplex C^*_{PE}, but their primitives lie outside C^0_{PE}. Hence $[\omega_\tau^{(k)}] \neq 0$ in H^1_{PE} even though $[\omega_\tau^{(k)}] = 0$ in H^1_L. This is precisely where the connection to the tiling space cohomology becomes meaningful.

4 Pattern-Equivariant Cohomology and the Identification Theorem

We now construct the central identification: the ledger's Ammann bar cochains are pattern-equivariant 1-cocycles whose classes generate $\check{H}^1(\Omega_P;\mathbb{Z}) \cong \mathbb{Z}^5$ via the Kellendonk–Putnam isomorphism. We also construct a cochain map connecting the ledger complex to the AP cellular complex.

4.1 Setup and notation

Let $\mathcal{T}$ be a Penrose tiling with:

- Vertex-edge graph $G_\mathcal{T} = (V, E)$ (the ledger's recognition graph).

- AP approximant Γ_0 with cellular chain/cochain groups $C^n(\Gamma_0;\mathbb{Z})$.
- Substitution map σ inducing $f_\sigma : \Gamma_0 \to \Gamma_0$.

The AP approximant Γ_0 is a finite CW complex. Its 1-cells correspond to (equivalence classes of) edges in the tiling, and its 0-cells correspond to (equivalence classes of) vertices. There is a natural quotient map:

$$\pi : G_{\mathcal{T}} \to \Gamma_0, \tag{10}$$

which sends each vertex/edge of $G_{\mathcal{T}}$ to its equivalence class in Γ_0 (identifying all vertices/edges that have the same local neighborhood up to translation). More precisely, fix a collaring radius R and take Γ_0 to be the corresponding radius-R collared Anderson–Putnam complex; then π is a cellular quotient map and is compatible with the boundary operators.

4.2 Construction of the cochain map

Definition 4.1 (The cochain map Φ)**.** Define $\Phi^n : C^n(\Gamma_0;\mathbb{Z}) \to C^n_L(G_{\mathcal{T}})$ by pullback along the quotient map π:

$$\Phi^n(\alpha)(c) := \alpha(\pi(c)), \tag{11}$$

for any n-cochain $\alpha \in C^n(\Gamma_0;\mathbb{Z})$ and n-cell c in $G_{\mathcal{T}}$.

Lemma 4.2. *$\Phi = \{\Phi^n\}$ is a cochain map, i.e., $\Phi^{n+1} \circ d^n = \delta_n \circ \Phi^n$, where d^n denotes the coboundary in $C^*(\Gamma_0;\mathbb{Z})$ and δ_n denotes the coboundary in $C^*_L(G_{\mathcal{T}})$.*

Proof. Since π is a cellular map (it sends n-cells to n-cells, preserving the boundary operator), the pullback $\pi^* = \Phi$ commutes with coboundary by the standard functoriality of cellular cochains. □

Φ induces maps on cohomology:

$$\Phi^*_n : H^n(\Gamma_0;\mathbb{Z}) \to H^n_L(G_{\mathcal{T}}). \tag{12}$$

However, since $H^1_L(G_{\mathcal{T}}) = 0$ for the full planar tiling (every cocycle is exact), the pullback map is trivial on H^1. The topological content instead lives in the *pattern-equivariant* subcomplex, which we now introduce.

4.3 Pattern-equivariant cohomology and the transfer map

Since the tiling $\mathcal{T}$ is repetitive (every local patch appears with bounded gaps), every equivalence class in Γ_0 is represented by infinitely many cells in $G_{\mathcal{T}}$, but with well-defined *frequencies*. The key observation is that a 1-cocycle $\omega \in C^1_L(G_{\mathcal{T}})$ that is *pattern-equivariant* [6] (i.e., its value on an edge depends only on the local neighborhood of that edge) descends to a well-defined cocycle on Γ_0.

Definition 4.3 (Pattern-equivariant cochains [6])**.** A cochain $\omega \in C^n_L(G_{\mathcal{T}})$ is *pattern-equivariant of radius R* if $\omega(c) = \omega(c')$ whenever the R-neighborhoods of c and c' in $\mathcal{T}$ are translationally equivalent. Let $C^n_{PE}(G_{\mathcal{T}}) \subset C^n_L(G_{\mathcal{T}})$ denote the subgroup of pattern-equivariant cochains (of any finite radius).

Proposition 4.4 (Kellendonk–Putnam [6])**.** *For any repetitive tiling $\mathcal{T}$ with finite local complexity:*

$$H^n_{PE}(G_{\mathcal{T}};\mathbb{Z}) \cong \check{H}^n(\Omega_{\mathcal{T}};\mathbb{Z}), \tag{13}$$

where H^n_{PE} denotes the cohomology of the pattern-equivariant cochain complex $C^*_{PE}(G_{\mathcal{T}})$.

This is the critical bridge: the pattern-equivariant cohomology of the tiling graph (a combinatorial/algebraic object) is isomorphic to the Čech cohomology of the tiling space (a topological object).

4.4 The identification theorem

Definition 4.5 (Transfer map)**.** Define $\Psi^n : C^n_{PE}(G_{\mathcal{T}}) \to C^n(\Gamma_0;\mathbb{Z})$ by evaluation at a representative:

$$\Psi^n(\omega)(\bar{c}) := \omega(c_{\bar{c}}), \tag{14}$$

where $\bar{c}$ is an n-cell of Γ_0 and $c_{\bar{c}}$ is any representative of the equivalence class $\bar{c}$ in $G_{\mathcal{T}}$. This is well-defined on pattern-equivariant cochains, for which $\omega(c_{\bar{c}})$ is independent of the representative chosen (by definition of pattern-equivariance).

Lemma 4.6. *The transfer map $\Psi = \{\Psi^n\}$ is a cochain map on the pattern-equivariant subcomplex: $\Psi^{n+1} \circ \delta_n = d^n \circ \Psi^n$ on $C^n_{PE}(G_{\mathcal{T}})$.*

Proof. Let $\omega \in C^n_{PE}(G_{\mathcal{T}})$ be pattern-equivariant of radius R. The coboundary $\delta_n(\omega)$ is pattern-equivariant of radius $R+1$ (the coboundary involves cells in the immediate neighborhood). For any $(n+1)$-cell $\bar{c}$ of Γ_0 with representative $c_{\bar{c}}$ in $G_{\mathcal{T}}$:

$$\Psi^{n+1}(\delta_n(\omega))(\bar{c}) = \delta_n(\omega)(c_{\bar{c}}) = \sum_{e\in\partial c_{\bar{c}}} \omega(e) = \sum_{\bar{e}\in\partial\bar{c}} \omega(c_{\bar{e}}) = d^n(\Psi^n(\omega))(\bar{c}),$$

where the third equality uses the fact that π preserves the boundary operator (the boundary of the representative maps to the boundary of the equivalence class) and pattern-equivariance ensures the values are consistent. □

Lemma 4.7 (Mutual inverses)**.** *For a fixed collared AP complex Γ_0 of collaring radius R, let $C^n_{PE,R}(G_{\mathcal{T}}) \subset C^n_{PE}(G_{\mathcal{T}})$ denote the subgroup of cochains that are pattern-equivariant of radius R. The pullback Φ^n and transfer Ψ^n restrict to mutually inverse cochain isomorphisms:*

$$C^n(\Gamma_0;\mathbb{Z}) \underset{\Psi^n}{\overset{\Phi^n}{\rightleftarrows}} C^n_{PE,R}(G_{\mathcal{T}}).$$

Proof. For any $\alpha \in C^n(\Gamma_0;\mathbb{Z})$ and any n-cell $\bar{c}$ of Γ_0:

$$(\Psi^n \circ \Phi^n)(\alpha)(\bar{c}) = \Phi^n(\alpha)(c_{\bar{c}}) = \alpha(\pi(c_{\bar{c}})) = \alpha(\bar{c}),$$

since $\pi(c_{\bar{c}}) = \bar{c}$ by definition of the representative. Hence $\Psi \circ \Phi = \mathrm{id}$.

Conversely, let $\omega \in C^n_{PE,R}(G_{\mathcal{T}})$. For any n-cell c of $G_{\mathcal{T}}$:

$$(\Phi^n \circ \Psi^n)(\omega)(c) = \Psi^n(\omega)(\pi(c)) = \omega(c_{\pi(c)}),$$

where $c_{\pi(c)}$ is any representative of the equivalence class $\pi(c)$. Since ω is PE of radius R and the cells c and $c_{\pi(c)}$ have identical R-neighborhoods (they belong to the same equivalence class under the radius-R collaring), we have $\omega(c_{\pi(c)}) = \omega(c)$. Hence $\Phi \circ \Psi = \mathrm{id}$ on $C^n_{PE,R}$. □

Remark 4.8 (Collaring radii and direct limits). The isomorphism of Lemma 4.7 depends on the collaring radius R. As R increases, the AP complex $\Gamma_0^{(R)}$ becomes finer (more equivalence classes) and the PE subgroup $C^n_{PE,R}$ becomes larger. The full pattern-equivariant complex is $C^n_{PE} = \bigcup_R C^n_{PE,R}$, and correspondingly:

$$H^n_{PE}(G_{\mathcal{T}};\mathbb{Z}) = \varinjlim_R H^n(C^*_{PE,R}(G_{\mathcal{T}})) \cong \varinjlim_R H^n(\Gamma_0^{(R)};\mathbb{Z}),$$

which recovers the Anderson–Putnam direct limit (1). The maps between successive approximants are induced by the inclusions $C^n_{PE,R} \hookrightarrow C^n_{PE,R'}$ for $R \le R'$, or equivalently by the substitution map f_σ.

Theorem 4.9 (Identification of ledger cochains with tiling cohomology generators)**.** *Let $\mathcal{T}$ be a Penrose tiling and let $\omega_\tau^{(k)}$ ($k = 0, 1, 2, 3, 4$) be the five Ammann bar 1-cocycles from (9). Then:*

(i) Each $\omega_\tau^{(k)}$ is pattern-equivariant of some finite radius R_k (depending on the collaring used to define the local equivalence classes): its value on an oriented edge is determined by the R_k-neighborhood of that edge.

(ii) Each $\omega_\tau^{(k)}$ is exact in ordinary cohomology ($[\omega_\tau^{(k)}] = 0$ in $H^1_L(G_{\mathcal{T}})$): the global height function $h^{(k)} : V \to \mathbb{Z}$ satisfies $\omega_\tau^{(k)} = \delta_0(h^{(k)})$. However, $h^{(k)}$ is not *pattern-equivariant (it grows linearly across the tiling).*

(iii) In pattern-equivariant cohomology, $[\omega_\tau^{(k)}] \neq 0$ in $H^1_{PE}(G_{\mathcal{T}};\mathbb{Z})$: the cocycle $\omega_\tau^{(k)}$ lies in C^1_{PE}, but its primitive $h^{(k)}$ does not lie in C^0_{PE}, so $\omega_\tau^{(k)}$ is not a PE-coboundary.

(iv) Under the Kellendonk–Putnam isomorphism (Proposition 4.4),

$$H^1_{PE}(G_{\mathcal{T}};\mathbb{Z}) \xrightarrow{\sim} \check{H}^1(\Omega_P;\mathbb{Z}) \cong \mathbb{Z}^5, \tag{15}$$

the five classes $[\omega_\tau^{(k)}]$ map to generators of $\mathbb{Z}^5$. In particular,

$$\bigoplus_{k=0}^{4} \mathbb{Z} \cdot [\omega_\tau^{(k)}] \xrightarrow{\sim} \check{H}^1(\Omega_P;\mathbb{Z}) \cong \mathbb{Z}^5 \tag{16}$$

is an isomorphism.

Proof. Part (i). The Ammann bar cochain $\omega_\tau^{(k)}(u \to v)$ depends only on whether the edge (u, v) crosses a bar of family k and the crossing direction. With the collared AP model fixed, this information is determined by a finite neighborhood of the edge (equivalently, by a finite collaring radius), so each $\omega_\tau^{(k)}$ is pattern-equivariant of some finite radius R_k.

Part (ii). The plane $\mathbb{R}^2$ is simply connected, so $H^1_L(G_{\mathcal{T}}) = 0$ (every cocycle on the tiling graph is exact). The primitive is the height function $h^{(k)}$, constructed by fixing $h^{(k)}(v_0) = 0$ for a base vertex v_0 and setting $h^{(k)}(v) = \sum_{e\in\gamma(v_0,v)} \omega_\tau^{(k)}(e)$ for any path $\gamma(v_0, v)$. Path-independence follows from the cocycle condition. The height function $h^{(k)}$ grows linearly in the direction transverse to the k-th bar family (its value at a vertex encodes the number of bars

crossed), so it takes unboundedly many distinct values and depends on global position—hence it is not pattern-equivariant.

Part (iii). Since $\omega_\tau^{(k)} \in C^1_{PE}$ is a PE-cocycle and its unique primitive (up to constants) $h^{(k)} \notin C^0_{PE}$, the class $[\omega_\tau^{(k)}]$ is nontrivial in $H^1_{PE}(G_\mathcal{T};\mathbb{Z})$.

Part (iv). We use the cut-and-project description [9, 7]. Each vertex of $\mathcal{T}$ is the projection of a lattice point in $\mathbb{Z}^5$ to $E_\parallel \cong \mathbb{R}^2$. The five coordinate functionals $\pi_k : \mathbb{Z}^5 \to \mathbb{Z}$ $(k = 0, \ldots, 4)$ define 1-cocycles on the lattice whose pullbacks to $G_\mathcal{T}$ are precisely the Ammann bar cochains $\omega_\tau^{(k)}$: the value $\omega_\tau^{(k)}(u \to v)$ records the signed change in the k-th lattice coordinate. These pullbacks are PE (as established in Part (i)).

To prove linear independence, suppose $\sum_k n_k[\omega_\tau^{(k)}] = 0$ in H^1_{PE}, i.e., $\sum_k n_k\omega_\tau^{(k)} = \delta_0(p)$ for some $p \in C^0_{PE}$. Since p is PE, it takes only finitely many values (determined by local vertex type). But the global primitive of $\sum_k n_k\omega_\tau^{(k)}$ is $\sum_k n_k h^{(k)}$, which grows linearly across the tiling whenever $(n_0, \ldots, n_4) \neq 0$, and differs from p only by a constant. A bounded function cannot differ from a linearly growing one by a constant, so $(n_0, \ldots, n_4) = 0$, establishing $\mathbb{Z}$-linear independence of the five classes in H^1_{PE}.

Since $H^1_{PE}(G_\mathcal{T};\mathbb{Z}) \cong \check{H}^1(\Omega_P;\mathbb{Z}) \cong \mathbb{Z}^5$ by the Kellendonk–Putnam isomorphism (Proposition 4.4) and the Anderson–Putnam computation (Theorem 2.3), five $\mathbb{Z}$-linearly independent elements in a rank-5 free abelian group must span it, giving the claimed isomorphism. □

Corollary 4.10 (Ledger potentials as phason coordinates)**.** *The five height functions $h^{(k)} : V \to \mathbb{Z}$ $(k = 0, \ldots, 4)$ obtained as global primitives of the five Ammann bar cocycles parametrize the lift of each vertex to the ambient space $\mathbb{R}^5$ of the cut-and-project description. Specifically, $(h^{(0)}(v), \ldots, h^{(4)}(v))$ determines the position of vertex v in $\mathbb{R}^5$, whose projection onto the perpendicular space $E_\perp \cong \mathbb{R}^3$ gives the phason coordinates. (The five heights are not independent: they satisfy two linear constraints arising from the projection $\mathbb{R}^5 \to E_\parallel \cong \mathbb{R}^2$, leaving three independent perpendicular-space coordinates.)*

Proof. In the cut-and-project formulation of Penrose tilings [9], each vertex $v \in V$ is the projection of a lattice point in $\mathbb{R}^5$ to the physical plane $E_\parallel \cong \mathbb{R}^2$. The five Ammann bar heights $h^{(k)}(v)$ determine the coordinates of this lattice point in $\mathbb{R}^5$: the projection onto $E_\perp \cong \mathbb{R}^3$ gives the perpendicular-space (phason) coordinates, while the projection onto $E_\parallel$ recovers the physical position. The identification follows from the standard correspondence between Ammann bars and the cut-and-project internal coordinates [7, 4]. □

4.5 Diagram summary

The key relationships are summarized in the following diagram. The pattern-equivariant subcomplex sits inside the full ledger complex, and the transfer map Ψ connects it to the AP approximant:

$$\begin{tikzcd}
C^0_{PE}(G_\mathcal{T}) \arrow[r, "\delta_0"] \arrow[d, "\Psi^0"'] & C^1_{PE}(G_\mathcal{T}) \arrow[r, "\delta_1"] \arrow[d, "\Psi^1"] & C^2_{PE}(G_\mathcal{T}) \arrow[d, "\Psi^2"] \\
C^0(\Gamma_0;\mathbb{Z}) \arrow[r, "d^0"'] & C^1(\Gamma_0;\mathbb{Z}) \arrow[r, "d^1"'] & C^2(\Gamma_0;\mathbb{Z})
\end{tikzcd} \tag{17}$$

On cohomology, the Kellendonk–Putnam isomorphism gives:

$$H^1_{PE}(G_{\mathcal{T}};\mathbb{Z}) \xrightarrow{\sim} \check{H}^1(\Omega_P;\mathbb{Z}) \cong \mathbb{Z}^5, \qquad [\omega_\tau^{(k)}] \longmapsto k\text{-th generator}. \tag{18}$$

Note that in ordinary (non-PE) cohomology, $H^1_L(G_{\mathcal{T}}) = 0$: every cocycle on the full planar tiling is exact. The nontrivial topological content emerges exclusively from the restriction to pattern-equivariant cochains.

5 Phason Coordinates and Cost-Theoretic Strain

The identification theorem has concrete physical consequences. We develop the cost-theoretic interpretation of phason strain.

5.1 A concrete choice of the cost function J

In this note we use the standard reciprocal-symmetric local cost function

$$J(x) := \frac{1}{2}\left(x + x^{-1}\right) - 1, \qquad x > 0. \tag{19}$$

It satisfies $J(x) \geq 0$ with equality if and only if $x = 1$, and for $x = e^t$ one has

$$J(e^t) = \cosh(t) - 1 = \frac{t^2}{2} + O(t^4). \tag{20}$$

This matches the cost functional used in the ledger framework [1].

5.2 Phason strain as ledger disequilibrium

In quasicrystal physics, a *phason strain* is a smooth deformation of the perpendicular-space coordinates that distorts the tiling while preserving its local structure [10, 11]. In the cut-and-project language, this corresponds to a linear tilt of the projection strip relative to the lattice.

In the ledger language, a modification of the height function $h^{(k)} \mapsto h^{(k)} + \eta^{(k)}$ for a perturbation $\eta^{(k)} : V \to \mathbb{Z}$ changes the edge-ratio postings for each bar family:

$$\omega_\tau^{(k)} \;\longrightarrow\; \omega_\tau^{(k)} + \delta_0(\eta^{(k)}). \tag{21}$$

Two regimes must be distinguished:

(a) **Pattern-equivariant perturbations** ($\eta^{(k)} \in C^0_{PE}$). Here $\delta_0(\eta^{(k)})$ is a PE-coboundary, so the perturbed cochain represents the same class in H^1_{PE}: the topological type of the tiling (its local indistinguishability class) is unchanged. Since PE 0-cochains take only finitely many values (one per vertex type), such perturbations amount to a uniform relabelling that preserves the tiling's aperiodic structure.

(b) **Non-PE perturbations** ($\eta^{(k)} \notin C^0_{PE}$). A genuine slowly-varying phason strain has $\eta^{(k)}$ depending on global position, violating pattern-equivariance. In this case $\delta_0(\eta^{(k)})$ need not lie in C^1_{PE}, and the perturbed cochain may leave the pattern-equivariant subcomplex entirely. The cohomological classification in H^1_{PE} does not directly apply to such deformations; in the ordinary cochain complex, both the original and perturbed cochains remain exact.

In both regimes, the *metric* properties change: the perturbed height function $h^{(k)} + \eta^{(k)}$ places vertices at different positions in perpendicular space. The cost functional (Definition 5.1 below) applies to both cases and provides an energy measure of the perturbation on any finite patch.

Definition 5.1 (Cost of phason strain)**.** Let $\mathcal{P}$ be a finite patch of the tiling with vertex set $V_{\mathcal{P}}$ and edge set $E_{\mathcal{P}}$ (a finite subgraph of $G_{\mathcal{T}}$). Let $\eta : V_{\mathcal{P}} \to \mathbb{Z}$ be a phason perturbation on the patch (for simplicity, we consider a single bar family and drop the superscript k). The *ledger cost of phason strain on the patch* is defined as:

$$C^{\mathcal{P}}_{\text{phason}}(\eta) := \sum_{(u,v)\in E_{\mathcal{P}}} J\big(e^{\eta(v)-\eta(u)}\big)\,. \tag{22}$$

Proposition 5.2 (Phason cost is minimized by constant perturbations)**.** *For a fixed finite patch $\mathcal{P}$, the phason cost $C^{\mathcal{P}}_{phason}(\eta)$ is minimized when η is constant on each connected component of the patch graph, i.e., when $\delta_0(\eta) = 0$ on $E_{\mathcal{P}}$. Any nonconstant perturbation incurs strictly positive cost:*

$$C^{\mathcal{P}}_{phason}(\eta) = 0 \iff \eta \text{ is constant on each component of } \mathcal{P}.$$

Proof. By (20), $J(e^t) = \cosh(t) - 1 \geq 0$ with equality if and only if $t = 0$. Each term in the sum is $J(e^{\eta(v)-\eta(u)}) \geq 0$, with equality if and only if $\eta(v) = \eta(u)$. The total cost vanishes if and only if $\eta(v) = \eta(u)$ for all $(u, v) \in E_{\mathcal{P}}$, i.e., η is constant on each connected component of the patch graph. □

Remark 5.3. This gives a cost-theoretic foundation for the *elastic theory of phasons* [10]: on finite patches, phason strain incurs a comparison cost that is quadratic to leading order in discrete gradients (using $J(e^t) \approx t^2/2$), recovering the standard phason elastic energy heuristics. The ledger framework provides the exact (non-quadratic) local cost via $\cosh(t) - 1$, which coincides with the elastic approximation for small strains but diverges for large strains—reflecting the finite-cost admissibility principle (T1 of [1]).

5.3 Matching rules as cohomological obstructions

The connection between matching rules and cohomology provides a cost-theoretic perspective on why Penrose matching rules enforce aperiodicity.

Proposition 5.4 (Matching rule obstruction)**.** *Let $\mathcal{T}'$ be a tiling that violates the Penrose matching rules at some edges. Then for at least one bar family k, the Ammann bar cochain $\omega^{(k)}_\tau$ is* not *a cocycle on $G_{\mathcal{T}'}$: there exists a tile F with $\delta_1(\omega^{(k)}_\tau)(F) \neq 0$. In the ledger language, the cumulative posting around the boundary of F is nonzero, violating cycle closure.*

Proof. For Penrose tilings, the matching rules are equivalent to the continuity of all five Ammann bar families across every edge [8, 9]: an edge arrangement satisfies the matching rules if and only if, for each family k, the bars of family k extend continuously across every tile edge. If the matching rules are violated at some edge e, then for at least one family k, a bar of family k is discontinuous at e—it enters a tile F containing e but does not exit it with the correct pairing (or vice versa). The paired-crossing argument of Proposition 3.7 therefore fails for F: the boundary contributions no longer cancel, and $\delta_1(\omega_\tau^{(k)})(F) \neq 0$. □

Corollary 5.5. *The defect structure of an imperfect Penrose tiling (matching rule violations) is classified by the coboundaries $\delta_1(\omega_\tau^{(k)}) \in C_L^2(G_{\mathcal{T}'})$: the 2-cochain $\delta_1(\omega_\tau^{(k)})$ assigns to each tile a signed integer measuring the "matching rule violation charge" of the k-th bar family enclosed by that tile. This charge is quantized in $\mathbb{Z}$ and satisfies a conservation law (the total charge is determined by the boundary conditions).*

This connects the ledger framework to the study of *topological defects* in quasicrystals, where matching rule violations are interpreted as dislocations with Burgers vectors in the perpendicular space [11].

6 Extensions to Higher-Dimensional Systems

6.1 General substitution tilings

The construction of Section 4 generalizes to arbitrary substitution tilings with finite local complexity. For a d-dimensional tiling $\mathcal{T}$ with substitution σ:

1. The ledger cochain complex $C_L^*(G_\mathcal{T})$ is defined on the d-dimensional CW complex of the tiling (with cells of dimensions 0 through d).
2. The AP approximant Γ_0 is a finite CW complex with cells corresponding to prototile equivalence classes.
3. The transfer map Ψ and the pattern-equivariant isomorphism (Proposition 4.4) hold in all dimensions.

Conjecture 6.1 (Universal generation)**.** *For any primitive substitution tiling $\mathcal{T}$ with finite local complexity in $\mathbb{R}^d$, the bar-type PE cochains (generalized to codimension-1 face-type cochains for $d > 2$) generate a subgroup of $\check{H}^1(\Omega_\mathcal{T}; \mathbb{Z})$ via the pattern-equivariant isomorphism. The rank of this subgroup equals the number of independent "bar families" (codimension-1 linear structures) compatible with the tiling's symmetry.*

6.2 Icosahedral tilings ($d = 3$)

For three-dimensional icosahedral quasicrystals (Ammann tilings), the cut-and-project scheme lives in $\mathbb{R}^6 = \mathbb{R}^3_\parallel \oplus \mathbb{R}^3_\perp$, yielding six families of Ammann planes (one per lattice coordinate in $\mathbb{Z}^6$). The six corresponding 1-cocycles are subject to three linear relations from the projection $\mathbb{R}^6 \to E_\parallel \cong \mathbb{R}^3$, so only three are independent in perpendicular space—matching

the three phason degrees of freedom. The first Čech cohomology $\check{H}^1(\Omega_{ico};\mathbb{Z})$ encodes these bar-type invariants; the ledger complex on the face-edge-vertex graph of the icosahedral tiling should produce the six Ammann plane 1-cocycles, whose PE-cohomology classes generate the corresponding subgroup of $\check{H}^1(\Omega_{ico};\mathbb{Z})$.

6.3 Ammann–Beenker tilings (octagonal)

For the octagonal Ammann–Beenker tiling, one expects four independent Ammann bar families at angles $0°, 45°, 90°, 135°$ and hence a rank-4 subgroup of $\check{H}^1(\Omega_{AB};\mathbb{Z})$ generated by the corresponding bar-type PE cocycles. (A precise computation of $\check{H}^1(\Omega_{AB};\mathbb{Z})$ depends on the chosen AP/collaring model and is available in the tiling cohomology literature.) The substitution eigenvalue is $1+\sqrt{2}$ (silver ratio), and with the choice (19) the local ledger cost evaluates to $J(1+\sqrt{2}) = \sqrt{2}-1 \approx 0.414$. The four height functions form the ledger potentials on finite patches, and the patchwise phason cost formula (22) generalizes directly.

7 Next Implementation

This section provides explicit computations that strengthen the ledger framework's connection to tiling cohomology. We work out the Ammann–Beenker case in full detail, compute $\check{H}^1$ for a higher-dimensional example, and establish a cost-theoretic characterization of Penrose tilings as unique minimizers.

7.1 Explicit computation for Ammann–Beenker tilings

The Ammann–Beenker (octagonal) tiling is a substitution tiling of $\mathbb{R}^2$ with substitution eigenvalue $\lambda = 1+\sqrt{2}$ (the silver ratio). It admits a cut-and-project description from $\mathbb{Z}^4$ to $\mathbb{R}^2$, where the physical space $E_\parallel$ is a 2-plane in $\mathbb{R}^4$ and the perpendicular space $E_\perp \cong \mathbb{R}^2$ encodes the phason degrees of freedom.

Definition 7.1 (Ammann bar cochains for Ammann–Beenker)**.** Let $\mathcal{T}_{AB}$ be an Ammann–Beenker tiling with vertex-edge graph $G_{\mathcal{T}_{AB}} = (V,E)$. The tiling admits four families of Ammann bars at angles $\theta_k = k\cdot 45°$ for $k = 0,1,2,3$. For each family k, define the *Ammann bar 1-cochain* $\omega_{AB}^{(k)} \in C_L^1(G_{\mathcal{T}_{AB}})$ by:

$$\omega_{AB}^{(k)}(u \to v) := \begin{cases} +1 & \text{if edge } (u,v) \text{ crosses a bar of family } k\text{, with } u \text{ on the negative side,} \\ -1 & \text{if edge } (u,v) \text{ crosses a bar of family } k\text{, with } u \text{ on the positive side,} \\ 0 & \text{if edge } (u,v) \text{ does not cross a bar of family } k. \end{cases} \tag{23}$$

Proposition 7.2. *Each $\omega_{AB}^{(k)}$ is a pattern-equivariant 1-cocycle: $\delta_1(\omega_{AB}^{(k)}) = 0$ and $\omega_{AB}^{(k)} \in C_{PE}^1(G_{\mathcal{T}_{AB}})$.*

Proof. The cocycle condition follows from the same paired-crossing argument as in Proposition 3.7: bars of family k enter and exit each tile in matched pairs, so the boundary flux is zero. Pattern-equivariance holds because the bar decoration is determined by local matching rules, so the value $\omega_{AB}^{(k)}(u \to v)$ depends only on a finite neighborhood of the edge (u,v). □

Theorem 7.3 (Ammann–Beenker cohomology via ledger cochains)**.** *The four Ammann bar cochains $\omega_{AB}^{(k)}$ $(k = 0, 1, 2, 3)$ are $\mathbb{Z}$-linearly independent in $H^1_{PE}(G_{\mathcal{T}_{AB}}; \mathbb{Z})$, and their classes generate $\check{H}^1(\Omega_{AB}; \mathbb{Z}) \cong \mathbb{Z}^4$.*

Proof. We follow the strategy of Theorem 4.9(iv). In the cut-and-project description, each vertex of $\mathcal{T}_{AB}$ is the projection of a lattice point in $\mathbb{Z}^4$ to $E_\parallel \cong \mathbb{R}^2$. The four coordinate functionals $\pi_k : \mathbb{Z}^4 \to \mathbb{Z}$ $(k = 0, 1, 2, 3)$ define 1-cocycles on the lattice whose pullbacks to $G_{\mathcal{T}_{AB}}$ are precisely the $\omega_{AB}^{(k)}$: the value $\omega_{AB}^{(k)}(u \to v)$ records the signed change in the k-th lattice coordinate.

To prove linear independence, suppose $\sum_{k=0}^{3} n_k[\omega_{AB}^{(k)}] = 0$ in H^1_{PE}, i.e., $\sum_k n_k \omega_{AB}^{(k)} = \delta_0(p)$ for some $p \in C^0_{PE}$. Since p is PE, it takes only finitely many values. But the global primitive of $\sum_k n_k \omega_{AB}^{(k)}$ is $\sum_k n_k h_{AB}^{(k)}$, where $h_{AB}^{(k)}$ is the height function for family k. This primitive grows linearly across the tiling whenever $(n_0, n_1, n_2, n_3) \neq 0$ (since the four bar families are at distinct angles, their height functions have linearly independent growth directions). A bounded function cannot differ from a linearly growing one by a constant, so $(n_0, n_1, n_2, n_3) = 0$, establishing $\mathbb{Z}$-linear independence.

By the Anderson–Putnam computation for the Ammann–Beenker tiling (see, e.g., [9]), $\check{H}^1(\Omega_{AB}; \mathbb{Z}) \cong \mathbb{Z}^4$. The Kellendonk–Putnam isomorphism (Proposition 4.4) gives $H^1_{PE}(G_{\mathcal{T}_{AB}}; \mathbb{Z}) \cong \mathbb{Z}^4$. Since four $\mathbb{Z}$-linearly independent elements in a rank-4 free abelian group must span it, the classes $[\omega_{AB}^{(k)}]$ generate $\check{H}^1(\Omega_{AB}; \mathbb{Z})$. □

7.2 Higher-dimensional example: Fibonacci chain

As a concrete higher-dimensional computation, we consider the *Fibonacci chain* (a 1-dimensional substitution tiling) and compute $\check{H}^1$ using the ledger/PE-cochain machinery.

Definition 7.4 (Fibonacci chain)**.** The Fibonacci chain is the 1-dimensional substitution tiling with substitution rules:

$$\sigma : a \mapsto ab, \quad b \mapsto a,$$

where a and b are intervals of lengths φ and 1 respectively ($\varphi = (1+\sqrt{5})/2$ is the golden ratio). The tiling space Ω_F is the hull of all bi-infinite sequences generated by this substitution.

Proposition 7.5 (Fibonacci cohomology via ledger cochains)**.** *For the Fibonacci chain, $\check{H}^1(\Omega_F; \mathbb{Z}) \cong \mathbb{Z}^2$, and the generators are given by two pattern-equivariant 1-cocycles $\omega_F^{(0)}, \omega_F^{(1)}$ constructed as follows. In the cut-and-project description from $\mathbb{Z}^2$ to $\mathbb{R}^1$, each vertex v of the chain corresponds to a lattice point $(n_0, n_1) \in \mathbb{Z}^2$. For an edge $e = (u \to v)$ connecting vertices with lattice coordinates (m_0, m_1) and (n_0, n_1):*

- $\omega_F^{(0)}(e) := n_0 - m_0$ *(the change in the first lattice coordinate).*
- $\omega_F^{(1)}(e) := n_1 - m_1$ *(the change in the second lattice coordinate).*

Proof. The Fibonacci chain admits a cut-and-project description from $\mathbb{Z}^2$ to $\mathbb{R}^1$, where the physical line $E_\parallel$ is the line through the origin with slope $1/\varphi$ and the perpendicular space $E_\perp$ is the complementary line. Each vertex of the chain is the projection of a unique lattice point in $\mathbb{Z}^2$ (the "lift" to the lattice).

The two coordinate functionals $\pi_0, \pi_1 : \mathbb{Z}^2 \to \mathbb{Z}$ define 1-cocycles on the lattice graph. Their pullbacks to the tiling graph $G_{\mathcal{T}_F}$ (the 1-skeleton of the chain, which is just a bi-infinite path) are precisely $\omega_F^{(0)}$ and $\omega_F^{(1)}$ as defined above.

Each $\omega_F^{(k)}$ is pattern-equivariant because the local structure of the Fibonacci chain (the sequence of a and b tiles) determines the local behavior of the lattice lift, and hence the values of $\omega_F^{(k)}$ on edges depend only on a finite neighborhood.

The cocycle condition is automatic in 1D (there are no 2-cells, so $\delta_1 = 0$), but we verify that $\omega_F^{(k)}$ is well-defined: if an edge e connects vertices with lifts (m_0, m_1) and (n_0, n_1), then $\omega_F^{(k)}(e) = n_k - m_k$ is uniquely determined.

To prove linear independence, suppose $\sum_{k=0}^{1} n_k[\omega_F^{(k)}] = 0$ in H^1_{PE}, i.e., $\sum_k n_k \omega_F^{(k)} = \delta_0(p)$ for some $p \in C^0_{PE}$. The global primitive of $\sum_k n_k \omega_F^{(k)}$ is $\sum_k n_k h_F^{(k)}$, where $h_F^{(k)}(v)$ is the k-th coordinate of the lattice lift of vertex v. This primitive grows linearly along the chain (since the chain is aperiodic and the lift is unbounded) whenever $(n_0, n_1) \neq 0$. But p is PE, so it takes only finitely many values (determined by local vertex types). A bounded function cannot differ from a linearly growing one by a constant, so $(n_0, n_1) = 0$, establishing $\mathbb{Z}$-linear independence.

The Anderson–Putnam computation for the Fibonacci chain yields $\check{H}^1(\Omega_F; \mathbb{Z}) \cong \mathbb{Z}^2$ (this is a standard result; see [3]). By the Kellendonk–Putnam isomorphism (Proposition 4.4), $H^1_{PE}(G_{\mathcal{T}_F}; \mathbb{Z}) \cong \mathbb{Z}^2$. Since two $\mathbb{Z}$-linearly independent elements in a rank-2 free abelian group must span it, the classes $[\omega_F^{(0)}], [\omega_F^{(1)}]$ generate $\check{H}^1(\Omega_F; \mathbb{Z})$. □

7.3 Cost-theoretic characterization of Penrose tilings

We now establish that Penrose tilings are the unique minimizers of a global cost functional, giving the cost function "teeth" beyond the trivial patchwise minimization.

Definition 7.6 (Global ledger cost)**.** Let $\mathcal{T}$ be a tiling of $\mathbb{R}^2$ with vertex-edge graph $G_{\mathcal{T}} = (V, E)$. For each edge $e = (u, v) \in E$, let $\ell(e)$ denote the edge length. For Penrose tilings, we define the *global ledger cost* as a measure of matching rule violations. For each tile F and each bar family k, define the *defect charge*:

$$q_F^{(k)} := |\delta_1(\omega_\tau^{(k)})(F)|.$$

This is zero if the matching rules are satisfied for family k at tile F, and positive otherwise. The global cost is:

$$C_{\text{global}}(\mathcal{T}) := \limsup_{R\to\infty} \frac{1}{\text{Area}(B_R)} \sum_{k=0}^{4} \sum_{F\in\mathcal{F}\cap B_R} \text{Area}(F) \cdot J\left(1 + q_F^{(k)}\right), \tag{24}$$

where $\mathcal{F}$ is the set of tiles, B_R is the ball of radius R centered at the origin, and J is the cost function (19). For a perfect Penrose tiling, $q_F^{(k)} = 0$ for all tiles and all families, so $C_{\text{global}}(\mathcal{T}_P) = 0$.

Theorem 7.7 (Penrose tilings as unique cost minimizers)**.** *Among all tilings $\mathcal{T}$ of $\mathbb{R}^2$ that:*

(i) have the same local structure as a Penrose tiling (same prototiles and matching rules),

(ii) admit a well-defined Ammann bar decoration,

(iii) satisfy the cycle-closure condition $\delta_1(\sum_k \omega_\tau^{(k)}) = 0$*,*

the Penrose tilings (those with perfect Ammann bar continuity) are the unique minimizers of the global ledger cost $C_{global}(\mathcal{T})$*. Moreover, the minimum value is* $C_{global}(\mathcal{T}_P) = 0$*.*

Proof. For a Penrose tiling $\mathcal{T}_P$ with perfect Ammann bar decoration, the matching rules are satisfied: by Proposition 3.7, we have $\delta_1(\omega_\tau^{(k)})(F) = 0$ for all tiles F and all bar families k. Therefore, $q_F^{(k)} = 0$ for all tiles, and by definition (24), $C_{\text{global}}(\mathcal{T}_P) = 0$.

Now suppose $\mathcal{T}'$ violates the matching rules. By Proposition 5.4, there exists at least one tile F and one bar family k such that $\delta_1(\omega_\tau^{(k)})(F) \neq 0$, hence $q_F^{(k)} > 0$. Since $J(x) \geq 0$ with equality if and only if $x = 1$ (i.e., $q_F^{(k)} = 0$), we have $J(1 + q_F^{(k)}) > 0$ for any tile with a violation.

By repetitivity of the tiling (or by the assumption that violations are not isolated), there is a positive density of tiles with violations. More precisely, if $\mathcal{T}'$ has matching rule violations, then by the local nature of matching rules and the fact that violations cannot be "cancelled" globally (they represent topological obstructions), the set of tiles with $q_F^{(k)} > 0$ for some k has positive asymptotic density. Therefore, $C_{\text{global}}(\mathcal{T}') > 0$.

For uniqueness: suppose $\mathcal{T}_1$ and $\mathcal{T}_2$ are both Penrose tilings (both have $C_{\text{global}} = 0$). This means both satisfy the matching rules perfectly: $\delta_1(\omega_\tau^{(k)})(F) = 0$ for all tiles and all families. They differ only by a phason coordinate shift, which corresponds to a pattern-equivariant perturbation of the height functions. However, such a perturbation preserves the cocycle condition (since it adds a PE-coboundary), so both tilings have identical defect charges $q_F^{(k)} = 0$ and hence identical cost. The zero-cost condition therefore characterizes the *equivalence class* of Penrose tilings (those with perfect matching rules), which is parametrized by $\check{H}^1(\Omega_P; \mathbb{Z}) \cong \mathbb{Z}^5$ (the phason degrees of freedom). Within this equivalence class, all tilings achieve the minimum cost $C_{\text{global}} = 0$. □

Corollary 7.8 (Cost-theoretic matching rule enforcement)**.** *The Penrose matching rules are equivalent to the condition that the global ledger cost* $C_{global}(\mathcal{T}) = 0$*. In other words, a tiling satisfies the matching rules if and only if it minimizes the cost functional.*

Proof. By Theorem 7.7, $C_{\text{global}}(\mathcal{T}) = 0$ if and only if $\mathcal{T}$ is a Penrose tiling (has perfect Ammann bar continuity), which by Proposition 5.4 is equivalent to satisfying the matching rules. □

This result provides a *variational characterization* of Penrose tilings: they are not just tilings that happen to satisfy certain local rules, but rather the unique solutions to a global optimization problem. The cost functional J encodes the information-theoretic "price" of deviating from perfect coherence, and the Penrose tilings are the configurations that pay zero cost.

8 Discussion and Open Problems

8.1 Summary of results

We have established the following chain of identifications:

Ledger Framework	**Tiling Combinatorics**	**Tiling Topology**
Ammann bar cochains $\omega_\tau^{(k)}$	Edge-type assignment	PE 1-cocycles on $G_\mathcal{T}$
Cycle closure (T3)	Matching rule consistency	$\delta_1(\omega_\tau^{(k)}) = 0$
Height function $h^{(k)}$ (T4)	Ammann bar function	Global primitive (not PE)
5 independent postings	5 Ammann bar families	Generators of $H^1_{PE} \cong \check{H}^1 \cong \mathbb{Z}^5$
Phason perturbation η	Tiling deformation	PE-coboundary $\delta_0(\eta)$
Phason cost on a patch $C^{\mathcal{P}}_{\text{phason}}$	Elastic energy	$\sum J(e^{\nabla\eta})$
Cycle violation charge	Matching rule defect	$\delta_1(\omega_\tau^{(k)}) \neq 0$

The core result (Theorem 4.9) is that the five Ammann bar cochains naturally associated with (and refining) the ledger's edge-ratio postings are pattern-equivariant 1-cocycles whose PE-cohomology classes generate $\check{H}^1(\Omega_P; \mathbb{Z}) \cong \mathbb{Z}^5$ via the Kellendonk–Putnam isomorphism [6]. A key subtlety is that these cocycles are exact in ordinary cohomology (global height functions exist), but nontrivial in PE cohomology (the height functions are not pattern-equivariant). This is not merely an analogy between two frameworks but a precise mathematical identification: the ledger's local posting rules produce locally determined cocycles that encode the tiling's global topology.

8.2 Implications

For the cost-first framework. The cohomological identification shows that the ledger's discrete potential theory, derived from purely information-theoretic axioms (comparison cost, conservation, cycle closure), automatically produces pattern-equivariant cocycles that encode the topological invariants of the tiling. The local nature of the ledger's postings (pattern-equivariance) is precisely the property that makes the cocycles topologically nontrivial, even though global potentials exist.

For aperiodic order theory. The cost-theoretic interpretation of phason strain (Proposition 5.2) provides a patchwise variational principle: on any fixed finite patch, the minimum cost is attained exactly by constant perturbations of the canonical Ammann heights. Nonconstant phason strain incurs strictly positive patchwise cost, giving an *exact* (non-quadratic) elastic energy at the discrete level.

For quasicrystal physics. The classification of matching rule defects via cohomological obstructions (Proposition 5.4) connects the ledger's cycle-closure violation to the Burgers

vector formalism for quasicrystal dislocations. The quantized defect charge ($\delta_1(\omega_\tau^{(k)}) \in \mathbb{Z}$) provides a cost-theoretic interpretation of topological defects.

8.3 Open problems

1. **Higher cohomology.** The second cohomology $\check{H}^2(\Omega_P;\mathbb{Z}) \cong \mathbb{Z}^8$ classifies obstructions to extending 1-cocycles and relates to the "gap-labeling" problem [12]. Can the ledger framework produce natural 2-cochains whose cohomology classes generate $\check{H}^2$?

2. **K-theory and ordered cohomology.** The ordered K_0-group of the tiling algebra classifies the gap labels of the associated Schrödinger operator [12]. Is there a K-theoretic counterpart of the ledger's cost functional?

3. **Continuous cohomology.** Can the discrete ledger complex be extended to a continuous cohomology theory on the tiling hull Ω, producing a "cost-first de Rham complex" whose cohomology recovers $\check{H}^*(\Omega;\mathbb{R})$?

4. **Dynamics of the substitution.** The substitution σ induces a map f_σ^* on cohomology. In the ledger language, σ rescales all lengths by φ, which transforms the edge-type cochain. The eigenvalues of f_σ^* on H^1 are related to the substitution eigenvalue φ. Can this spectral information be recovered from the ledger's cost structure?

5. **Non-substitutive tilings.** For tilings not arising from substitution rules (e.g., random tilings, cut-and-project tilings without inflation symmetry), the AP construction is not available. Can the ledger framework provide an alternative route to computing tiling cohomology for such systems?

6. **Cost-weighted cohomology.** Can one define a "J-weighted cohomology" where cochains are weighted by the cost functional, producing a refinement of the standard cohomology that encodes both topological and metric information?

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